

INSTITUTE OF ACCOUNTANCY ARUSHA



Department of Computer Science And Mathematics

ONLINE INDUSTRIAL PRACTICAL TRAINING APPLICATION SYSTEM

A Project Proposal Submitted in Partial Fulfilment of the Requirements
for the Assignment of Bachelor of Science in Cybersecurity

Submitted by:

Sn	STUDENT NAME	REGISTRATION NUMBER
1	HEMED R. SAID	BCSE-01-0493-2025
2	SUBIRA M. WILLIAM	BCSE-01-0052-2025
3	DAVID A. KATISHA	BCSE-01-0187-2025
4	ANGLE J. MTEI	BCSE-01-0018-2025
5	MINIS P. SIMBEYE	BCSE-01-0231-2025

05/2026

Declaration

We, hereby declare that the work presented in this project proposal titled "Online IPT Application System" is our own original work carried out under the supervision of Dr. Kiwango. The work has not been previously presented in part or full to any other institution for the award of any academic qualification.

I further declare that this proposal complies with the academic integrity and ethical guidelines of Institute Of Accountancy Arusha (IAA)

ACKNOWLEDGMENT

First and foremost, We give glory and honor to the Almighty God for granting us the wisdom, strength, and perseverance to undertake this academic work. His grace and guidance have been our source of strength throughout this journey.

We are profoundly grateful to my supervisor, Dr. Kiwango, for the invaluable guidance, constructive criticism, and scholarly mentorship provided throughout the course of this project. Your dedication and patience have greatly shaped the quality of this work.

We extend our sincere appreciation to all academic staff of the Department of Computer Science and Mathematics at Institute Of Accountancy Arusha (IAA) for creating an enabling learning environment and for the knowledge imparted during our period of study.

Special thanks go to our colleagues and fellow students who offered peer support, shared ideas, and engaged in constructive discussions that enriched this research.

We also acknowledge the support of the Institute Of Accountancy Arusha (IAA) ICT department and library staff, whose resources and facilities were instrumental in the preparation of this proposal.

Finally, heartfelt gratitude goes to our families for their unending prayers, financial support, and encouragement. This achievement is as much yours as it is ours.

ABSTRACT

The Industrial Practical Training (IPT) programme is a critical component of undergraduate education that bridges the gap between theoretical academic knowledge and real-world professional practice. However, the conventional management of IPT application processes in many higher educational institutions remains largely manual, paper-based, and inefficient. These traditional approaches are riddled with challenges including slow processing of applications, loss of important documents, limited accessibility, lack of transparency, and difficulty in coordinating between students, departments, companies, and administrators.

This project proposes the design and development of an Online IPT Application System a secure, web-based platform that automates and streamlines the entire IPT application workflow. The proposed system will provide a centralized digital environment through which students can submit IPT applications online, upload required documents, select preferred placement organizations, and track the status of their applications in real time. Academic supervisors and departmental administrators will be able to review, approve, and manage applications efficiently through a dedicated administrative dashboard.

From a Cybersecurity perspective, the system will incorporate robust security mechanisms including role-based access control (RBAC), data encryption, secure user authentication (including multi-factor authentication), input validation, and audit logging to protect sensitive student and institutional data against unauthorized access and Cyber threats.

The proposed system is expected to significantly reduce administrative workload, improve turnaround time for application processing, enhance communication between all stakeholders, and ensure data integrity throughout the IPT cycle. The system will be developed using modern web technologies and hosted on a secure server infrastructure.

Keywords: Industrial Practical Training (IPT), Online Application System, Cybersecurity, Role-Based Access Control, Web-Based System, Secure Authentication, University Administration.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Industrial Practical Training (IPT), also known as Industrial Attachment, Work-Based Learning, or Internship, is a structured academic programme that enables university students to acquire practical, on-the-job experience in their respective fields of study. It is a mandatory component of most undergraduate programmes, particularly in science, technology, engineering, and mathematics (STEM) disciplines. The programme is designed to expose students to real industry environments, allowing them to apply theoretical knowledge, develop professional competencies, and build industry networks that are essential for post-graduation employment.

The management of the IPT programme involves multiple stakeholders, including students, academic supervisors, departmental coordinators, industry companies or host organisations, and university administrators. Coordinating these stakeholders requires an efficient communication and data management system. Unfortunately, a large number of universities, particularly in developing regions, continue to rely on manual, paper-based processes for managing IPT applications and placements.

In a typical manual system, students are required to physically collect application forms from departmental offices, fill them out by hand, attach supporting documents, and submit the paperwork to a coordinator for processing. The coordinator then manually verifies and forwards the applications to supervisors or host organisations. This process is not only time-consuming but also prone to errors, delays, loss of documents, and lack of transparency.

The rapid advancement of information and communication technologies (ICTs), particularly the proliferation of web-based applications and cloud computing, has opened up new possibilities for automating administrative processes in academic institutions. Web-based systems offer accessibility, scalability, and efficiency that far exceed what manual processes can achieve. Countries and institutions that have adopted digital management systems for their academic processes have reported significant improvements in operational efficiency, student satisfaction, and data management.

1.2 Overview of the Proposed System

The Online IPT Application System is a secure, role-based, web-accessible platform designed to automate and manage the end-to-end process of Industrial Practical Training application and placement at the university level. The system integrates modern web development technologies with established Cybersecurity principles to deliver a platform that is both user-friendly and secure.

The system is structured around four primary user roles: Students, Academic Supervisors, Departmental Administrators, and System Administrators. Each role is granted specific permissions and access rights defined through Role-Based Access Control (RBAC),

ensuring that users can only access information and perform actions relevant to their responsibilities.

Students will be able to register and log in securely, fill and submit IPT application forms online, upload supporting documents (such as letters of acceptance and academic transcripts), view available host organisations, and track the real-time status of their applications. Academic supervisors will review student applications, provide feedback, assign students to companies, and monitor student progress. Departmental administrators will manage application workflows, generate placement reports, and communicate with host organisations. System administrators will manage user accounts, system configurations, and security settings.

The system will employ secure HTTPS connections, password hashing, multi-factor authentication (MFA), encrypted data storage, comprehensive audit logs, and intrusion detection mechanisms to safeguard the platform against common Cybersecurity threats such as SQL injection, cross-site scripting (XSS), unauthorized access, and data breaches.

CHAPTER TWO

PROBLEM STATEMENT

The Industrial Practical Training (IPT) programme is an indispensable part of undergraduate academic training. However, the process of managing IPT applications in many universities remains heavily dependent on manual, paper-based workflows that are increasingly incompatible with the demands of modern academic administration. Several critical challenges undermine the effectiveness and security of the current system, as described below.

The following specific problems have been identified in the existing IPT application management process:

- i. **Inefficiency of Manual Processes:** The current system requires students to physically collect, complete, and submit paper-based application forms. This process introduces significant delays in processing times, as each application must be manually verified, reviewed, and forwarded through multiple layers of bureaucracy. The cumulative effect is a slow, resource-intensive workflow that places unnecessary administrative burdens on staff and frustrates students.
- ii. **Risk of Document Loss and Data Integrity Issues:** Physical documents are vulnerable to loss, damage, or misplacement. Students who rely on the manual system risk having their applications misplaced or lost entirely, with no digital record of submission. Furthermore, manual transcription of application data introduces the risk of human error, which can compromise data accuracy and integrity.
- iii. **Limited Accessibility and Inconvenience:** Students are required to be physically present at the institution to submit applications, access forms, and follow up on their application status. This requirement is particularly problematic for students who are on industrial attachment in remote locations or those who face challenges with transportation. The lack of remote accessibility creates inequity in access to administrative services.
- iv. **Lack of Real-Time Application Tracking:** In the current system, students have no reliable mechanism for tracking the status of their IPT applications in real time. Communication about application progress is largely informal, inconsistent, and often delayed, leaving students uncertain about the outcome of their submissions.
- v. **Poor Communication and Coordination:** The manual system lacks a structured communication channel between students, supervisors, departmental coordinators, and host organisations. This results in miscommunication, missed deadlines, duplicated efforts, and general inefficiency in the coordination of IPT placements.
- vi. **Absence of Cybersecurity Protections:** Paper-based systems provide no protection against unauthorized access to student data. Sensitive personal information, academic records, and placement details are handled without encryption, access controls, or audit trails. This exposes students and institutions to risks of identity theft, data manipulation, and privacy violations.

- vii. **Challenges in Record Keeping and Reporting:** The generation of accurate and timely reports on IPT applications, placements, and student performance is extremely difficult with the current manual system. Without a centralized digital repository, administrators cannot easily produce analytics, monitor trends, or make data-driven decisions about the IPT programme.

In summary, the existing manual IPT application process is characterized by inefficiency, lack of transparency, security vulnerabilities, and poor user experience. There is a compelling need for a secure, automated, and web-based solution that addresses all of these challenges while adhering to modern Cybersecurity standards.

CHAPTER THREE

OBJECTIVES OF THE STUDY

3.1 Main Objective

The main objective of this project is to design, develop, and implement a secure, web-based Online IPT Application System that automates and streamlines the Industrial Practical Training application and management process at the university, while incorporating comprehensive Cybersecurity controls to protect the confidentiality, integrity, and availability of all system data and user interactions.

3.2 Specific Objectives

In order to achieve the main objective, the following specific objectives will be pursued:

- i. To conduct a thorough analysis of the existing manual IPT application process, identifying specific operational inefficiencies, security vulnerabilities, and stakeholder pain points that the proposed system must address.
- ii. To design a secure, multi-role, web-based application system that supports distinct user roles, including students, academic supervisors, departmental administrators, and system administrators, each with appropriately scoped permissions enforced through Role-Based Access Control (RBAC).
- iii. To develop a student-facing module that enables online registration, secure login, digital submission of IPT application forms with supporting document uploads, selection of preferred host organisations, and real-time application status tracking.
- iv. To implement a supervisor and administrator dashboard that facilitates efficient review, approval, and management of IPT applications, generation of placement reports, and monitoring of student progress throughout the IPT cycle.
- v. To integrate robust Cybersecurity mechanisms into the system, including HTTPS encryption, multi-factor authentication (MFA), bcrypt password hashing, input validation and sanitization, SQL injection and XSS prevention, audit logging, and session management controls.
- vi. To design and implement a secure database schema with proper data normalization, referential integrity, and access restrictions to ensure the confidentiality and integrity of student, institutional, and placement data.
- vii. To conduct functional testing, security vulnerability assessments, and user acceptance testing (UAT) to validate that the system meets its functional requirements and adheres to established Cybersecurity best practices.
- viii. To evaluate the performance and usability of the deployed system by collecting feedback from target users and measuring improvements in application processing time, data accuracy, and user satisfaction compared to the existing manual process.

CHAPTER FOUR

SIGNIFICANCE OF THE PROJECT

The development and deployment of the Online IPT Application System is expected to generate far-reaching benefits across multiple stakeholder groups. These benefits extend from immediate operational improvements at the institutional level to broader implications for Cybersecurity practice and digital transformation in academic administration.

4.1 Benefits to Students

- i. **Convenience and Accessibility:** Students will be able to submit IPT applications and required documents from any location with internet access, eliminating the need for physical visits to the university during the application period. This is particularly beneficial for students in remote areas or those already on attachment.
- ii. **Real-Time Application Tracking:** The system provides students with instant visibility into the status of their applications, reducing uncertainty and enabling them to plan accordingly. Automated notifications will keep students informed of key milestones in the application process.
- iii. **Reduced Risk of Document Loss:** Digital submissions are stored securely in a centralized database, eliminating the risk of physical document loss or damage. Students can retrieve and resubmit documents as needed without having to start the process from scratch.
- iv. **Improved Communication:** The built-in messaging and notification system will enable direct, documented communication between students and their academic supervisors, facilitating timely feedback and guidance.

4.2 Benefits to the Institution

- i. **Operational Efficiency:** By automating repetitive administrative tasks, the system will significantly reduce the workload on departmental staff, enabling them to focus on higher-value activities such as quality assurance and student support.
- ii. **Centralized Data Management:** All IPT-related data will be stored in a structured, searchable database, enabling administrators to generate reports, identify trends, and make evidence-based decisions about the IPT programme.
- iii. **Institutional Image and Competitiveness:** The adoption of a modern, digital IPT management system signals the institution's commitment to innovation and quality education, enhancing its reputation among prospective students and partner organisations.
- iv. **Regulatory Compliance:** A well-documented, auditable system with proper data handling procedures will help the institution comply with data protection regulations and academic quality standards.

4.3 Benefits to Administrators and Supervisors

- i. **Streamlined Workflow:** Administrators will benefit from a structured, step-by-step workflow for processing applications, with automated notifications and task assignments reducing the risk of oversight or missed deadlines.
- ii. **Audit Trails and Accountability:** Every action performed within the system is logged with timestamps and user identifiers, providing a comprehensive audit trail that supports accountability and facilitates the investigation of any disputes or irregularities.
- iii. **Data-Driven Decision Making:** With access to real-time dashboards and historical reports, administrators can monitor IPT programme performance, identify bottlenecks, and implement improvements based on reliable data.

4.4 Contribution to Cybersecurity Practice

Beyond its operational benefits, this project makes a meaningful contribution to the field of applied Cybersecurity. By implementing and documenting a range of security controls including authentication, authorization, encryption, and vulnerability mitigation techniques, the project serves as a practical case study in the application of Cybersecurity principles within a real-world software development context. The documented design choices, threat models, and security testing outcomes will be valuable for future research and for Cybersecurity practitioners working in academic information systems.

CHAPTER FIVE

TECHNICAL GAPS

A review of the existing IPT management practices and comparable systems in similar institutions reveals a number of significant technical gaps. These gaps relate to both the limitations of the current manual process and the shortcomings of existing digital solutions where they have been partially implemented. The identification of these gaps is essential for informing the design and development of the proposed Online IPT Application System.

5.1 Absence of Secure Authentication and Access Control

Most existing manual systems and even some early digital implementations lack robust authentication mechanisms. Single-factor password-based authentication is insufficient in the face of modern credential-based attacks such as phishing, brute-force, and credential stuffing. Furthermore, the absence of role-based access control in manual processes means that sensitive data is often accessible to all staff, regardless of their actual responsibilities, violating the principle of least privilege.

5.2 Lack of Data Encryption and Secure Storage

In manual systems, student data, including personal identification, academic records, and placement information, is stored in physical files or, at best, unencrypted spreadsheets. Even some basic digital systems fail to implement database encryption or secure file storage, exposing sensitive data to unauthorized access. There is a clear gap in the application of data-at-rest and data-in-transit encryption standards.

5.3 Insufficient Input Validation and Injection Prevention

Many basic web-based alternatives that have been developed lack proper input validation and sanitization routines. This leaves them vulnerable to critical web application attacks such as SQL injection, cross-site scripting (XSS), and command injection, attacks that are consistently listed among the top vulnerabilities in the OWASP Top 10. The absence of these controls poses severe risks to the integrity and availability of the system.

5.4 No Audit Logging or Activity Monitoring

The current manual system maintains no systematic record of who accessed what information, when, or what changes were made. This absence of audit trails makes it impossible to detect insider threats, investigate data tampering, or maintain accountability. Even where digital tools are partially used, comprehensive logging is rarely implemented, representing a critical gap in operational security.

5.5 Limited Integration and Interoperability

Existing systems, where they exist are often soiled, operating independently without integration with the university's central student information system, email infrastructure, or

identity management platform. This lack of interoperability leads to data duplication, inconsistency, and additional administrative overhead. There is a technical gap in the design of integrated, API-driven systems that connect with the broader university IT ecosystem.

5.6 Poor Mobile Responsiveness and Accessibility

Many existing digital tools used for IPT administration are not optimized for mobile devices. Given that a significant proportion of students in developing regions access the internet primarily through smartphones, the failure to provide a mobile-responsive interface constitutes a significant accessibility gap that limits participation and adoption.

5.7 Absence of Vulnerability Assessment and Secure Development Practices

A recurring gap in existing systems is that they were developed without a formal Secure Software Development Life-cycle (SSDLC). Security was not considered during design, and penetration testing or vulnerability assessments were not conducted prior to deployment. This has resulted in systems that remain exposed to known vulnerabilities long after launch.

5.8 No Multi-Factor Authentication (MFA)

Most existing academic application systems rely solely on username and password for authentication. In an era of increasingly sophisticated Cyber-attacks targeting educational institutions, the absence of multi-factor authentication represents a significant gap in identity assurance, particularly for administrator accounts that have elevated system privileges.

CHAPTER SIX

PROPOSED SYSTEM

6.1 System Description

The Online IPT Application System is a secure, centralized, multi-tier web application designed to fully digitalize and manage the end-to-end Industrial Practical Training application life-cycle. The system is built upon a three-tier architecture comprising a presentation layer (front-end), an application logic layer (back-end server), and a data layer (secure relational database). All communications between tiers are encrypted using TLS/HTTPS protocols.

The system will be developed using established web technologies. The front-end will be built with HTML5, CSS3, and JavaScript (with a responsive framework such as Bootstrap or Tailwind CSS) to ensure cross-platform compatibility and mobile responsiveness. The back-end will be developed using a server-side language such as PHP or Python (Django/Flask), with a MySQL or PostgreSQL relational database for persistent data storage. The system will be hosted on a Linux-based web server (Apache/Nginx) with HTTPS enforced via SSL/TLS certificates.

6.2 System Users and Roles

The system defines four primary user roles, each with distinct permissions enforced by Role-Based Access Control (RBAC):

- i. Student: Can register and authenticate, complete and submit IPT application forms, upload supporting documents, view available host organisations, track application status, receive notifications, and communicate with supervisors.
- ii. Academic Supervisor: Can log in, view and review assigned student applications, approve or reject applications with comments, monitor student placement progress, and generate individual student reports.
- iii. Departmental Administrator: Can manage the full application workflow, approve final placements, communicate with host organisations, generate departmental placement statistics and reports, and manage the list of approved host organisations.
- iv. System Administrator: Can manage all user accounts and roles, configure system settings, monitor system logs and audit trails, perform security configurations, and carry out system maintenance tasks.

6.3 Key Features of the Proposed System

- i. Secure User Registration and Authentication: Students and staff register with verified institutional email addresses. All passwords are hashed using bcrypt. Multi-factor authentication (MFA) via time-based one-time passwords (TOTP) is available for all user roles, and mandatory for administrator accounts.
- ii. Digital Application Form: A structured, dynamic online application form replaces paper-based forms. The form includes fields for student personal details, academic information, preferred host organisations, proposed IPT period, and supporting documentation uploads (PDF/JPEG).
- iii. Document Upload and Management: Students can securely upload required documents (letters of acceptance, CV, academic transcripts) directly through the platform. Uploaded files are scanned for malware, stored in an encrypted file storage system, and accessible only to authorized users.
- iv. Application Tracking Dashboard: Both students and administrators have access to a real-time dashboard that displays application status (Submitted, Under Review, Approved, Rejected, Deferred). Automated email and in-app notifications are triggered at each stage change.
- v. Role-Based Dashboards: Each user role has a customized dashboard displaying only relevant information and action items, enforcing the principle of least privilege and improving the user experience.
- vi. Host Organization Database: A searchable directory of approved host organisations is maintained within the system, allowing students to select preferred placements. Administrators can add, update, or deactivate organization records.

- vii. **Communication Module:** An integrated messaging system allows students to communicate with supervisors and administrators within the secure platform, creating a documented, auditable communication trail.
- viii. **Reporting and Analytics:** Administrators can generate detailed reports on application volumes, placement rates, student distribution across industries, and processing times. Data can be exported in PDF or CSV formats.
- ix. **Audit Logging and Activity Monitoring:** All user actions, including logins, form submissions, approvals, and data modifications, are recorded in an immutable audit log with timestamps and IP addresses, supporting accountability and forensic investigation.
- x. **Security Controls:** The system implements HTTPS/TLS for all data transmission, prepared statements and parameterized queries to prevent SQL injection, output encoding to prevent XSS, CSRF tokens, session timeout controls, account lockout after failed login attempts, and regular automated security scans.

6.4 Expected Outcomes

- i. A fully functional, secure, and tested Online IPT Application System accessible via web browsers on desktop and mobile devices.
- ii. A reduction of not less than 70% in manual administrative effort associated with IPT application processing.
- iii. A measurable improvement in application processing turnaround time, from weeks to days.
- iv. A secure, centralized repository of all IPT application data with zero instances of document loss.
- v. A documented Cybersecurity framework for the system, including a threat model, implemented security controls, and a vulnerability assessment report.
- vi. Positive user acceptance testing (UAT) results demonstrating that the system meets functional requirements and is usable by all target user groups.

Overall, the Online IPT Application System will represent a significant advancement in the digital transformation of academic administration, demonstrating how Cybersecurity principles can be effectively integrated into the design and deployment of a practical, real-world web application.